LOADOUT: MACHINE LEARNING FOR ARMY LOGISTICS COST PREDICTION

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> “Amateurs talk about tactics, but professionals study logistics.”
>
> **General Robert H. Barrow**

1 Business Understanding

Loadout Overview:

Loadout replaces slow, manual Army movement request processes with a streamlined web app, allowing users to quickly submit details like inventory, personnel, and special needs for faster coordination and planning.

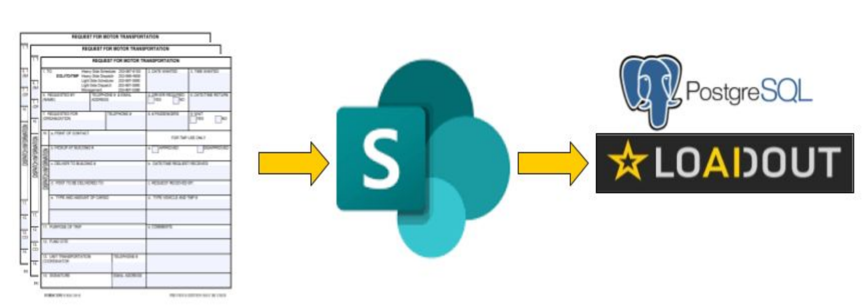
Motivation:

We aim to build a cost estimation service using Loadout’s existing data, giving users quick, data-driven cost predictions to accelerate decision-making.

Problem:

With limited available features at inference, how can we develop a reliable model to predict movement costs (USD) to support faster, more informed planning?

Army Logistics Tracking Mechanisms



PDF Forms	Sharepoint	Loadout Db
Structured Entry	Structured Entry	Structured Entry
Not Transparent	Transparent	Transparent
Not Customizable	Not Customizable	Customizable

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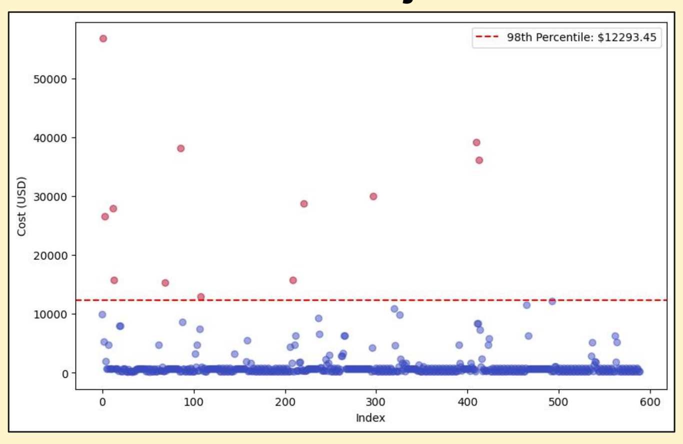
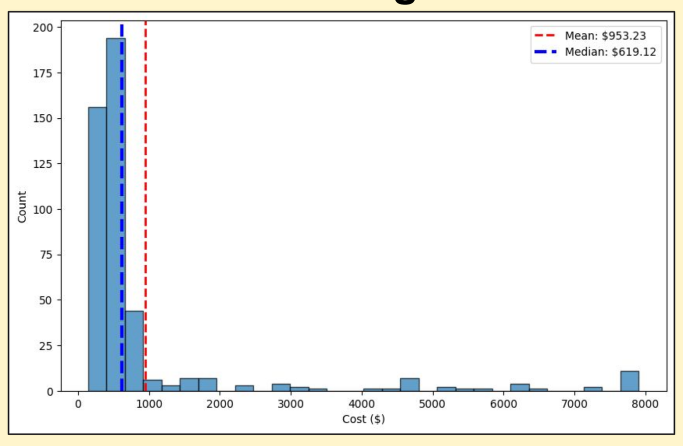
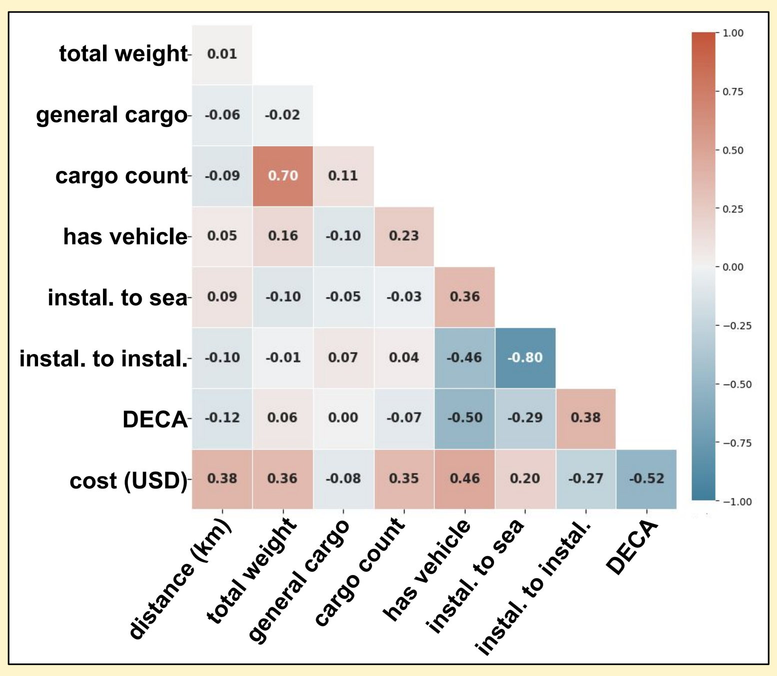
Cost Outlier Adjustment



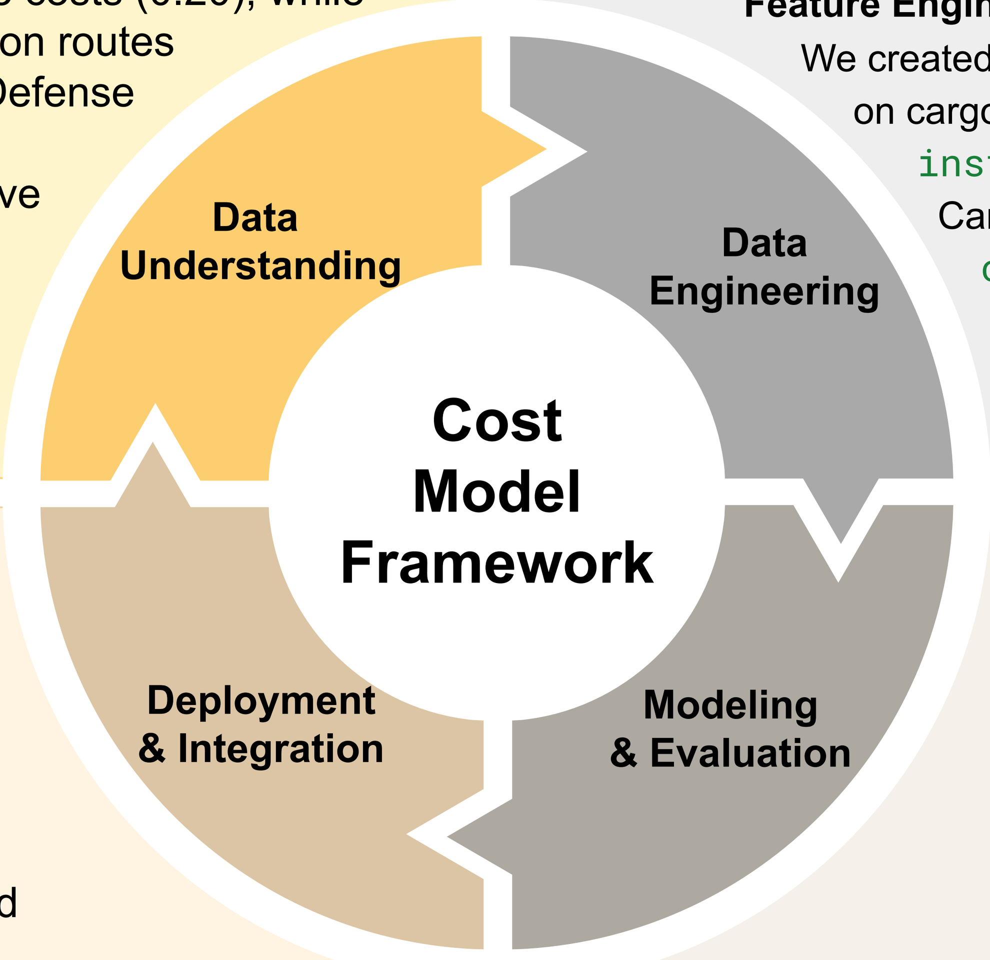
Cost Histogram



To evaluate predictive modeling feasibility in Loadout, we conducted an exploratory data analysis on transportation costs and durations. We cleaned and engineered key features, removed implausible records, and capped outliers to ensure data quality. This refined dataset enabled reliable model prototyping and highlighted the most predictive features.



The correlation matrix shows that distance (0.38), cargo weight (0.36), and vehicle shipments (0.46) are strong positive predictors of transportation costs. Installation-to-seaport routes slightly increase costs (0.20), while installation-to-installation routes reduce them (-0.27). Defense Commissary Agency (DECA) operations have the strongest negative correlation (-0.52), indicating consistent cost savings.



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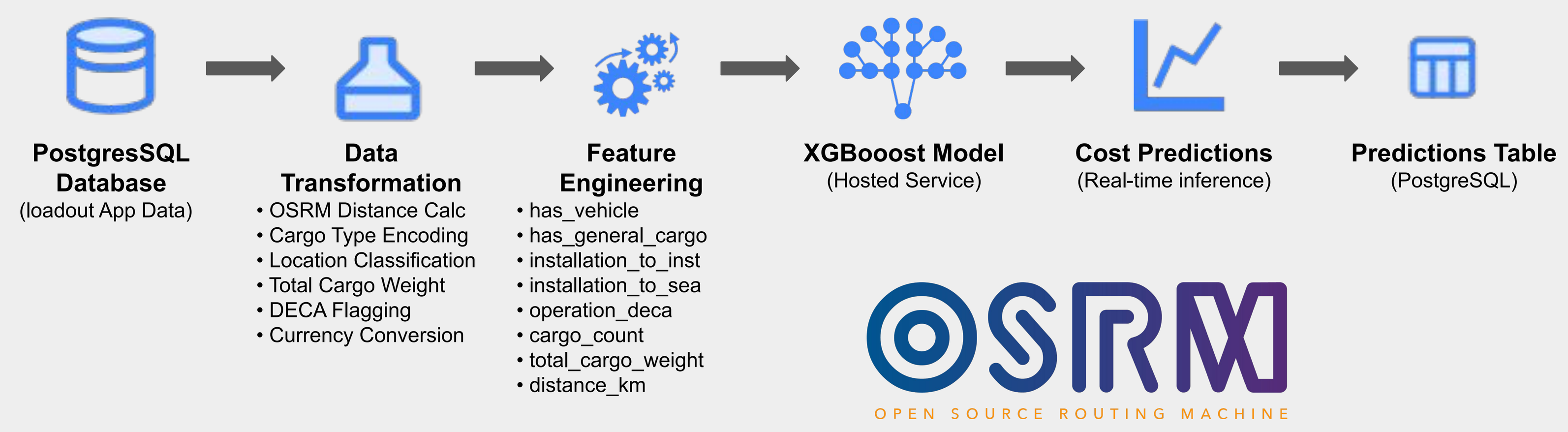
We built an end-to-end ETL pipeline to power Loadout’s predictive cost estimation feature for military logistics, using historical data from its PostgreSQL database. The pipeline transforms complex relational data into model-ready features and feeds them into an XGBoost inference system for real-time cost predictions. Key transformations included computing real-world distances with the OSRM API, detecting DeCA operations using wildcard logic, and normalizing costs across currencies using historical exchange rates.

LOADOUT PREDICTIVE COST ESTIMATION ETL PIPELINE



Feature Engineering Summary:

We created binary features like `has_vehicle` and `has_general_cargo` based on cargo type, and route classification features such as `installation_to_installation` and `installation_to_seaport`. Cargo weight was aggregated into `total_cargo_weight`, and `cargo_count` was retained as a key numeric feature. All transformed data was stored in a dedicated PostgreSQL predictions table to streamline model training and inference.

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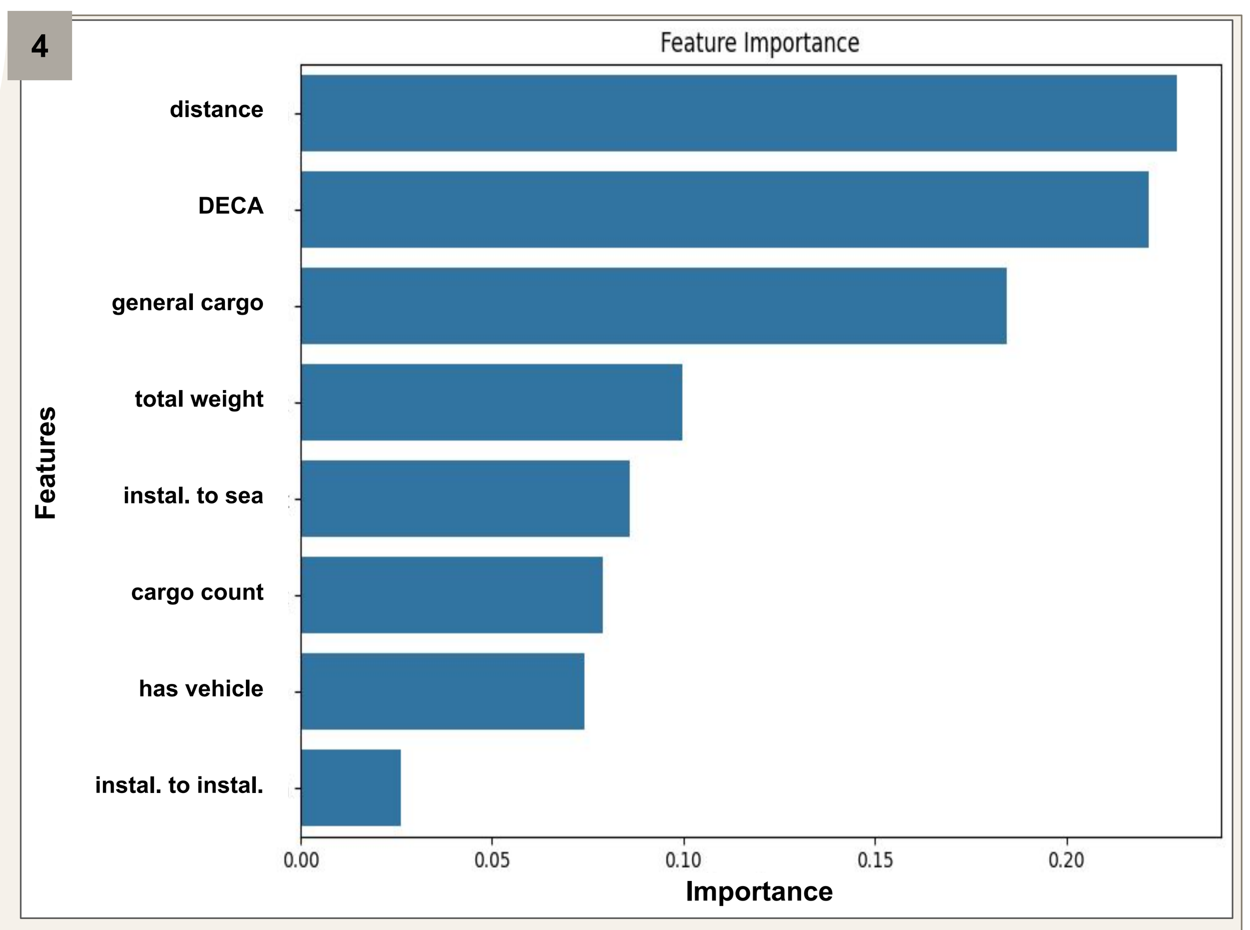
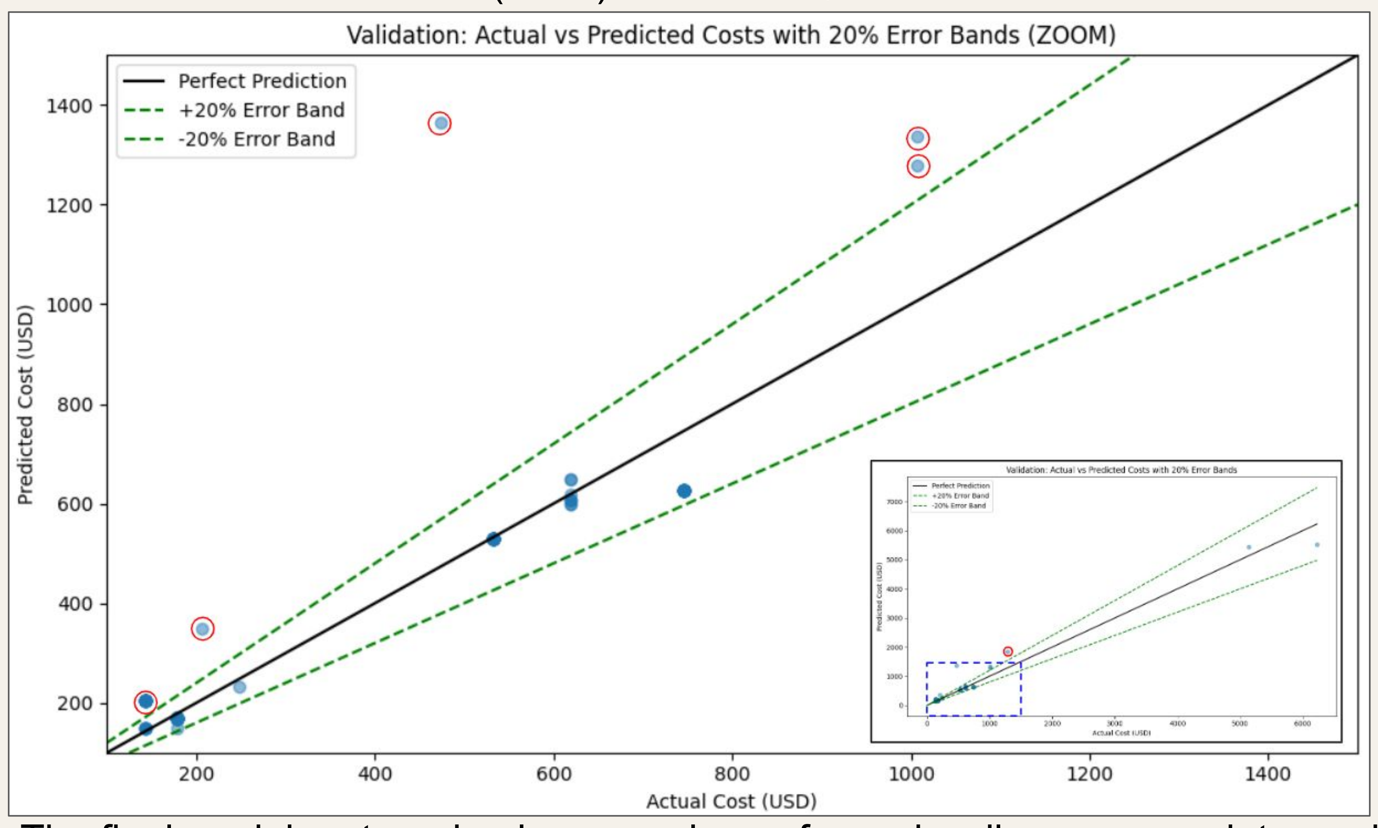


Our team evaluated eight machine learning algorithms to predict ground movement costs using Loadout data, ultimately selecting XGBoost for its robustness and strong performance on tabular data. We used an 80/20 train-test split on 455 requests, stratified the target variable into cost bins, and performed hyperparameter tuning via random search with 5-fold cross-validation. The model needed to meet the following business metrics:

1. Predicted cost should be within +/- 20% of the actual cost 95% of the time.
2. Mean Absolute Error (MAE) should be $100 or less.



Validation Performance Metrics:

Metric	Value
MAE	$89.22
RMSE	$585.92
20%	98.33%
R^2	0.3498
Time	0.0013s

The final model met our business goals, performed well on unseen data, and was containerized and deployed via a RESTful API within the Loadout environment for real-time cost prediction.

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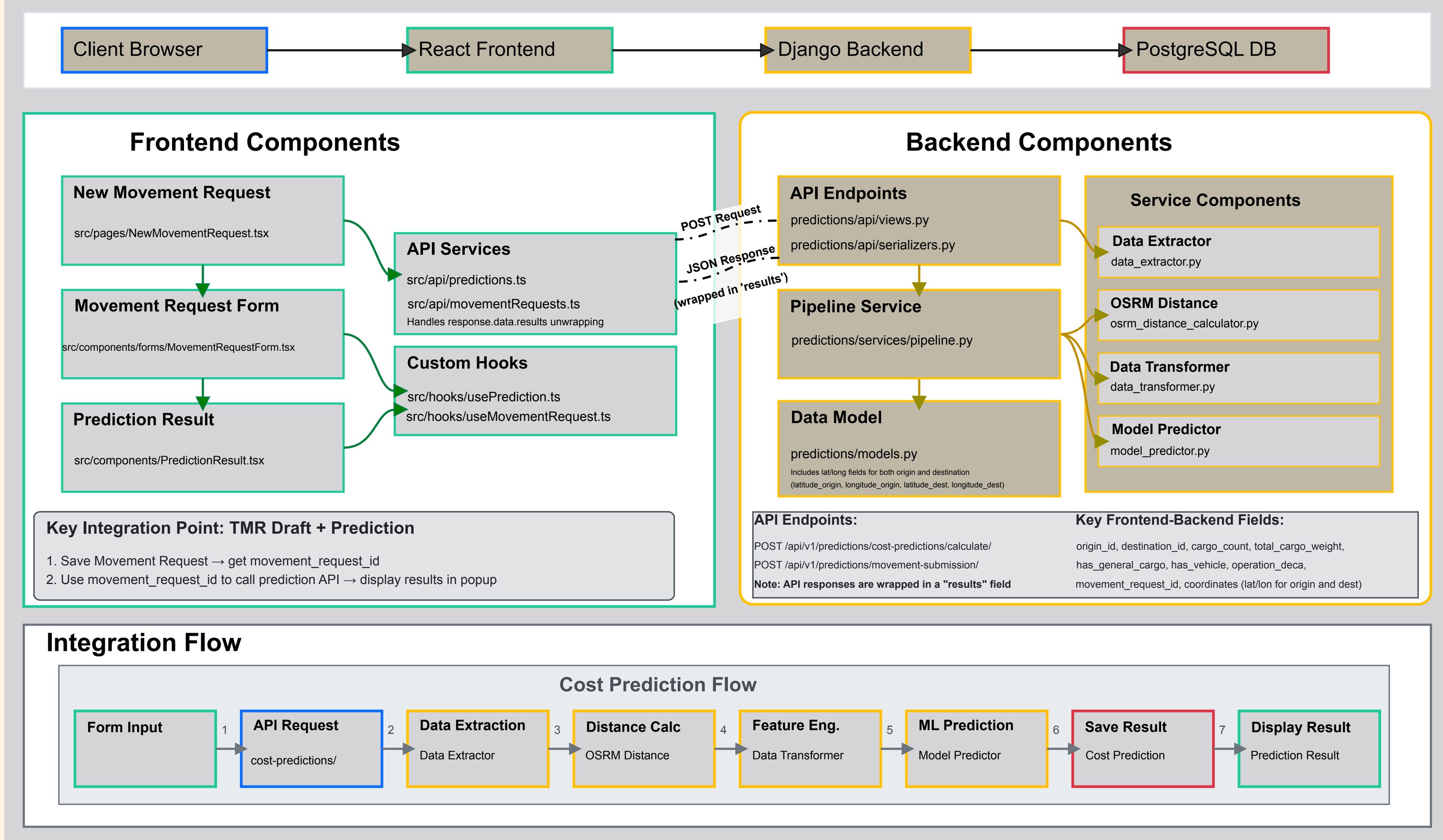
Deployment Architecture: Containerized FastAPI with pre-stored ML models on auto-scaling Kubernetes with CI/CD pipeline. Protected by input validation, structured error handling, comprehensive logging, and security controls.

Key Learnings: Cloud testing, scalable architecture, and continuous integration maintained both deployment agility and stakeholder satisfaction.



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QR Code: Link to additional resources.

Future Work

The successful deployment of the Loadout Capstone predictive service demonstrates our group’s ability to deliver a scalable, impactful solution. By leveraging Kubernetes, Docker, and CI/CD pipelines, we addressed technical and operational challenges, delivering a service that enhances the Loadout application’s functionality. Future work will focus on final testing, monitoring with Kibana, and collecting more data to refine and validate our predictive model.

AI Technicians 2024-2025 - Capstone Project - Supervisors: Ethan Cook, Anthony Fortuna